

DashAR: An Augmented Reality-based Heads Up Display for Automobiles

An Abstract Presentation by Trevor D. Brown

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Presentation Flow

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History

Automotive
Instrumentation and the
Heads Up Display.

02

Opportunity

A new approach to
Automotive HUDs.

03

Research

Prior investigations into
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Objectives

Building an AR-based HUD
Framework.

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Demonstration

Using an AR-based HUD
Framework.

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Next Steps

Where can (and will) this
research go next?

History

Automotive Instrumentation and the Heads Up Display (HUD)

History of Automotive Instrumentation



Ammeter

(1925)

Ford Model T

History of Automotive Instrumentation (continued)



Various Gauges

(2017)

Subaru Forester

History of Automotive Instrumentation (continued)

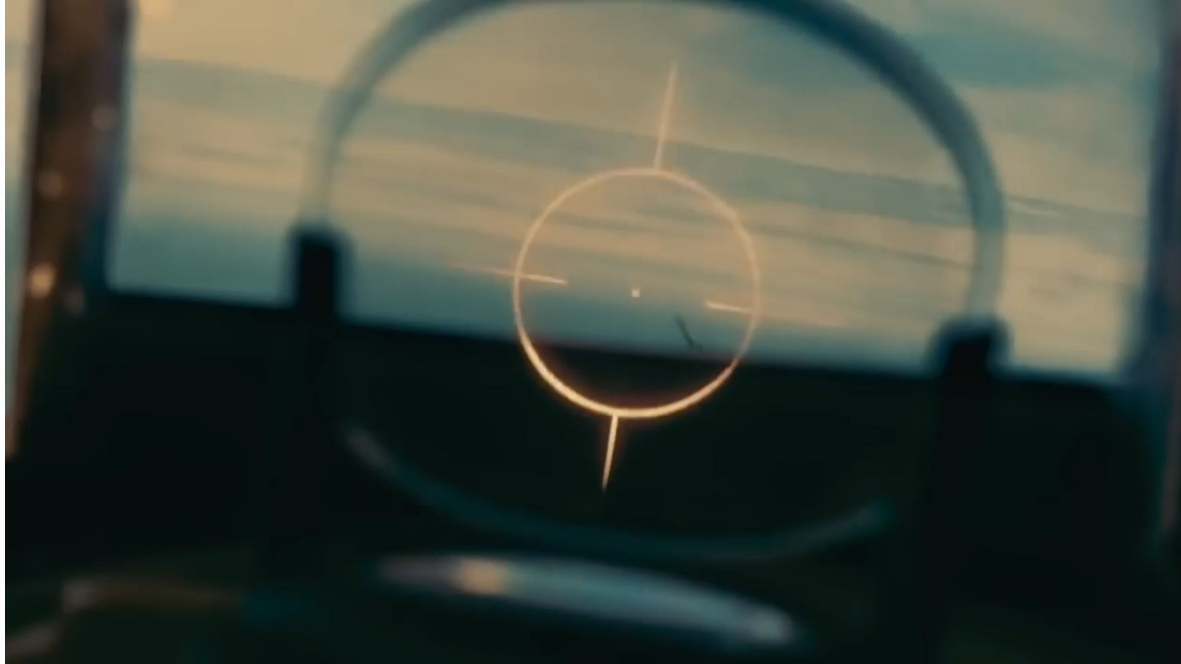


Various Gauges

(2017)

Subaru Forester

History of Heads Up Displays (HUDs)



Fighter Plane HUDs

(1942 - present)

Frame from *Dunkirk* (2017)

History of HUDs (continued)



Fighter Plane HUDs

(1999)

Lockheed Martin C-130J
Super Hercules

History of HUDs (continued)



Automobile HUDs

(1980 - present)

BMW E60

Opportunity

A new approach to Automotive HUDs.

In 300 feet, Turn Right onto Elm Street.
Lookout Point → Home

SW
230°



950
rpm

0
mph

48%
Fuel

D

Parking Brake Engaged



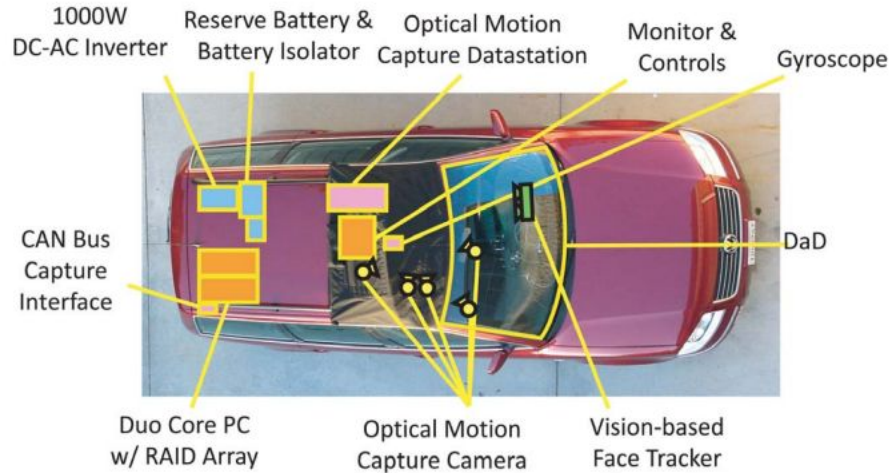
BRAKE



Research

Prior investigations into Automotive HUDs.

Literature Review



43/45



45 43

A. Doshi, S. Yuanhsien, et. al. (2009). *A Novel Active Heads-Up Display for Driver Assistance*. IEEE Transactions on Systems, Man, and Cybernetics - Part B.

- Windshield Implementation of a HUD while driving; alerts of adverse conditions.

Literature Review (continued)

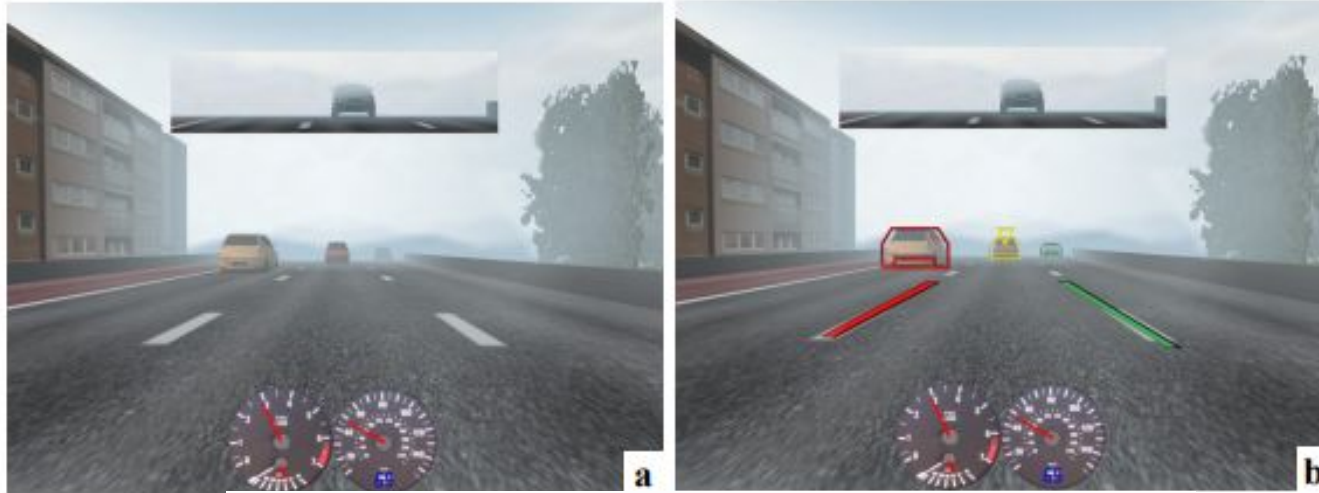
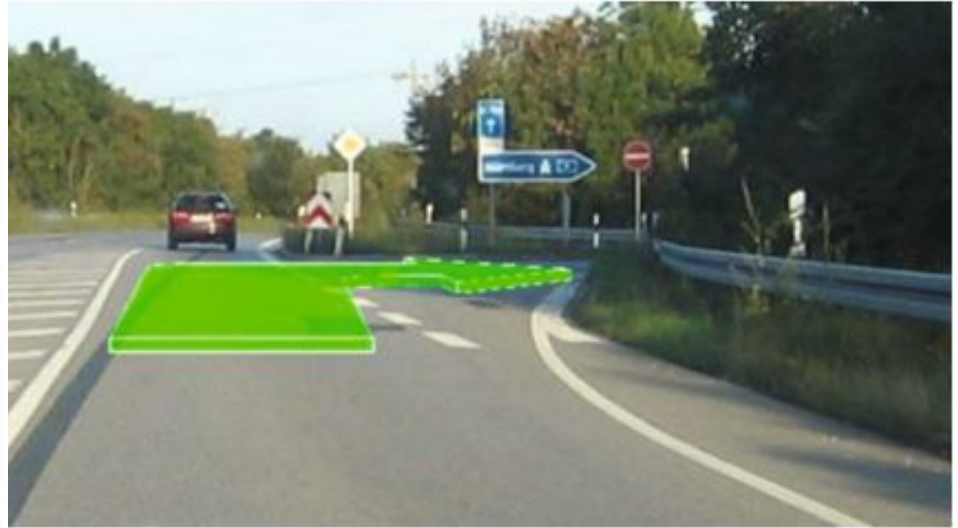


Figure 1: Driving under low visibility conditions
a) With contemporary HDD, and b) With proposed HUD visual aid

V. Charissis and S. Papanastasiou. (2006). *Human-Machine Collaboration Through Vehicle Head Up Display Interface*. European Annual Conference on Human Decision-Making and Manual Control.

- Heads Up vs. Heads Down Display. Consideration for low visibility conditions. Object Detection.

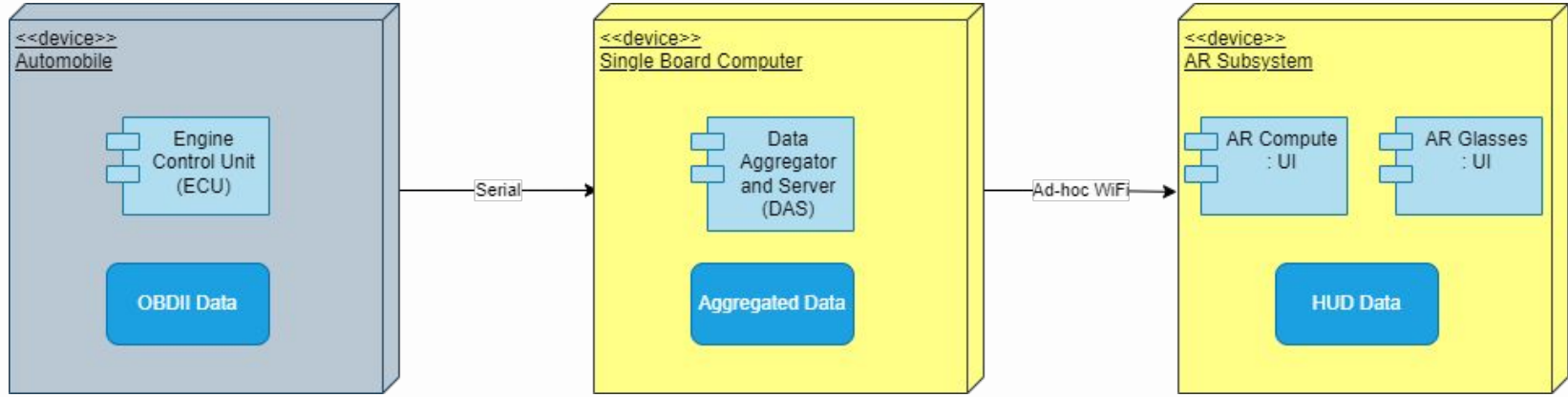
Literature Review (continued)



- A. Pauzie. (2015). *Head Up Display in Automotive: A New Reality for the Driver*. International Conference on Human-Computer Interaction.
 - Reasoning for improved “eyes on the road” time using HUD.

Objectives

Building an AR-based HUD Framework.



Development Approach

Data Aggregator and Server

Simple REST API to access a vehicle's OBDII interface, its related data, as well as data from the internet (third-party APIs, such as Google Maps.)

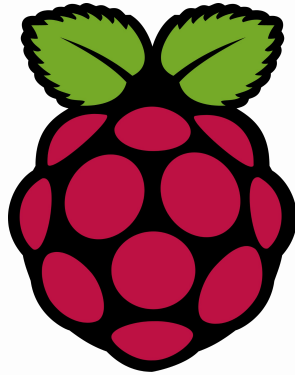
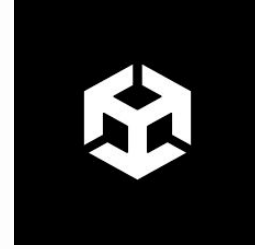
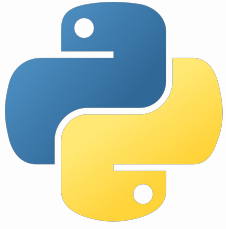
HUD

Tailored specifically for the XREAL line of AR glasses, HUD focuses on three levels of data: critical operation, non-critical operation, and amenities.

Administrative Functions

User customizable HUDs to meet the needs of end user, select which data needs to be retrieved, and on what frequency.

Development Tools



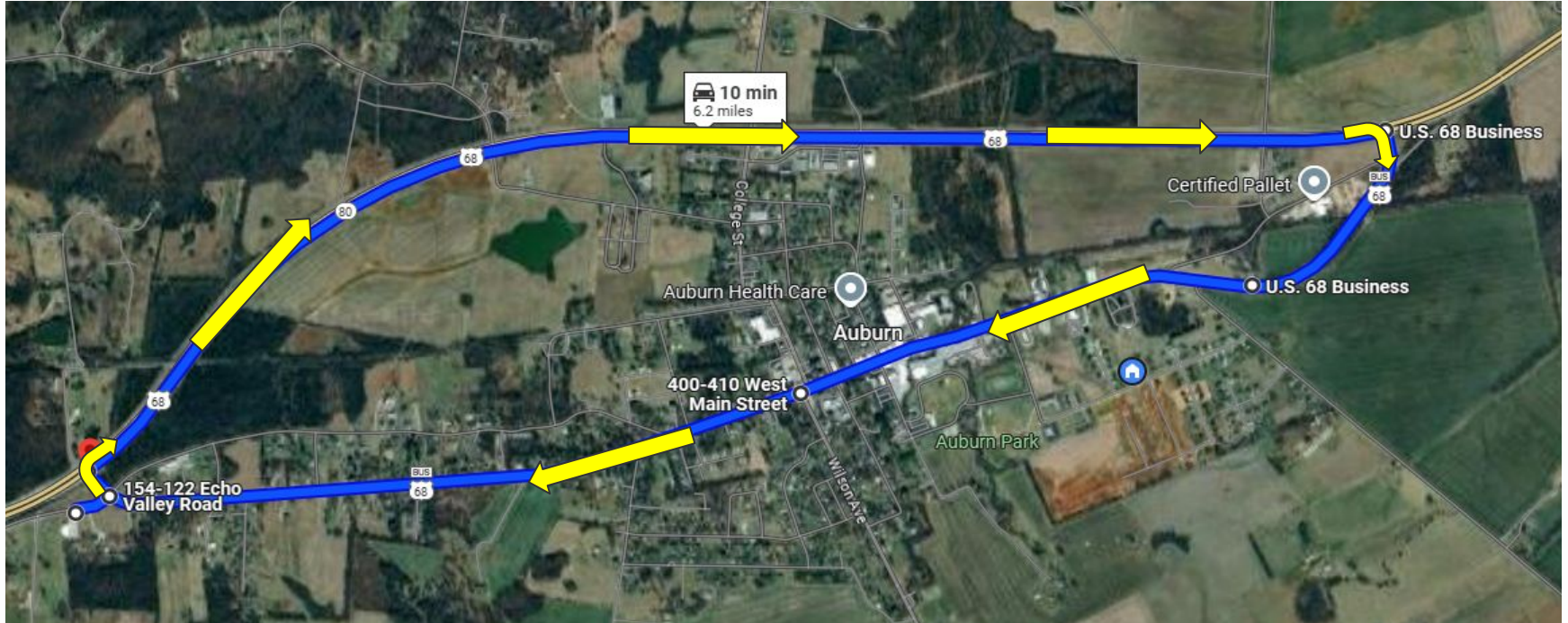
Project Schedule

[illegible]

Demonstration

Using an AR-based HUD.

Test Drive (11/10/2024)





Technical Challenges (thus far...)

Data Aggregator and Server

Safe power down (e.g. UPS, battery pack), establishing third-party API connectivity (e.g. cellular modem, hotspot)

HUD

Data request methodology (e.g. Client/Server (synchronous), Pub/Sub (asynchronous), etc.), non-distracting UI, dynamic UI through configurations, Restrictions to additional apps while driving.

UX

Intuitive and non-distracting, usability in any conditions (rain, snow, night, etc.)

Approximate Cost

Data Aggregator and Server

Raspberry Pi 5 Kit - \$160

12V to 120V Power Inverter (150W) - \$40

Surge Protector - \$10

HUD

XREAL Air 2 Pro - \$400

XREAL Beam Pro - \$200

Total Cost

\$ 810

Next Steps

Where can (and will) this research go next?

By Spring 2025...

Expanding the HUD's functionality

Support for third-party APIs (e.g. Google Maps Navigation, Weather Conditions), additional widgets (e.g. shifter position, instrumentation alerts, etc.)

Administrative Interface

Web interface, available on the Raspberry Pi, to customize the HUD, data retrieved, etc.

Testing with Human Subjects

Real-world testing, to elicit feedback on developments and their effectiveness.

Future Works

Operational Data

Advanced Instrumentation (for auto racers), Hauling Data (for truckers),
Customized Navigation (for military applications)

Intelligent Operation

Computer Vision applications (e.g. blind spot and object detection), Operating
Limit Monitoring and Alerts (e.g. unsafe vehicle speed and acceleration, excess
load weight when hauling, etc.)

Optimizations

Deeper integration into automobiles (e.g. CAN Bus)

References

- A. Doshi, S. Yuanhsien, et. al. (2009). *A Novel Active Heads-Up Display for Driver Assistance*. IEEE Transactions on Systems, Man, and Cybernetics - Part B.
- V. Charissis and S. Papanastasiou. (2006). *Human-Machine Collaboration Through Vehicle Head Up Display Interface*. European Annual Conference on Human Decision-Making and Manual Control.
- A. Pauzie. (2015). *Head Up Display in Automotive: A New Reality for the Driver*. International Conference on Human-Computer Interaction.
- G. Burnett (2003). *A road-based evaluation of a Head-Up Display for presenting navigation information*. Proceedings of HCI International Conferences.
- M. Lemley and E. Volokh (2018). *Law, Virtual Reality, and Augmented Reality*. University of Pennsylvania Law Review Vol. 166 No. 5.

Additional Sources

- XREAL Developer (<https://developer.xreal.com/>)
- Unity 3D (<https://unity.com>)
- Figma (<https://figma.com>)
 - Car Instrument Cluster Iconography
 - <https://www.figma.com/design/kqfyR7SKw7yCR5qKxzAlo2/>
- Python (<https://www.python.org/>)
- OBDII
(<https://learn.sparkfun.com/tutorials/obd-ii-uart-hookup-guide/obd-commands>)

Additional Sources (continued)

- Images
 - Dunkirk Fighter Plane HUD
 - <https://youtu.be/Z52vVsMdBkc?t=118>
 - Automotive HUD, Fighter Jet HUD
 - https://en.wikipedia.org/wiki/Head-up_display
 - Ford Model T
 - <https://modeltfordfix.com/the-1925-model-t-ford/>

Thanks!

Do you have any questions?

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